

Road Accident Data Analysis Report

Excel-based analysis using Power Query, Pivot Tables, Pivot Charts and dashboard techniques

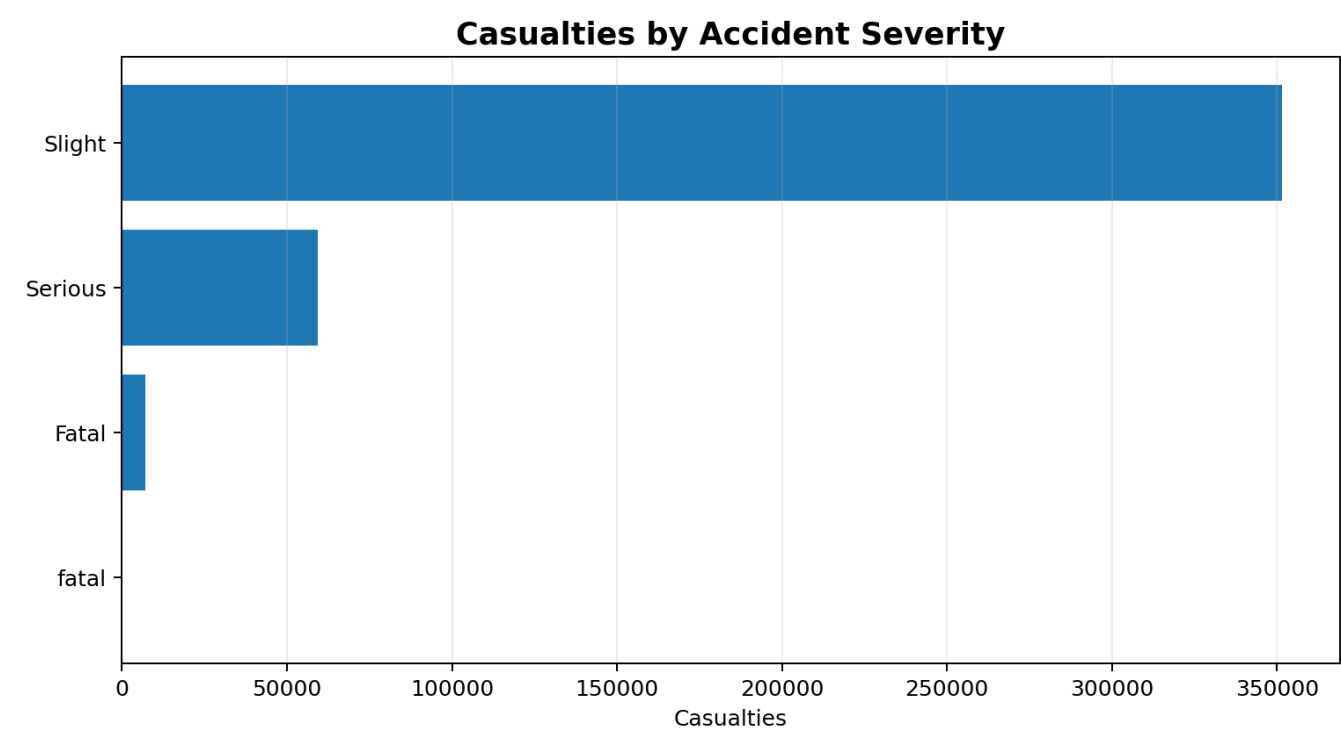
Metric	Value
Total accident records	307,973
Total casualties	417,883
Total vehicles involved	563,302

1. Executive Summary

The workbook contains 307,973 accident records and 417,883 reported casualties. The analysis uses Pivot Tables and Pivot Charts to summarize severity, vehicle type, road characteristics, lighting conditions and monthly trends. The results show where casualty volumes are concentrated and which road and environmental conditions deserve closer attention.

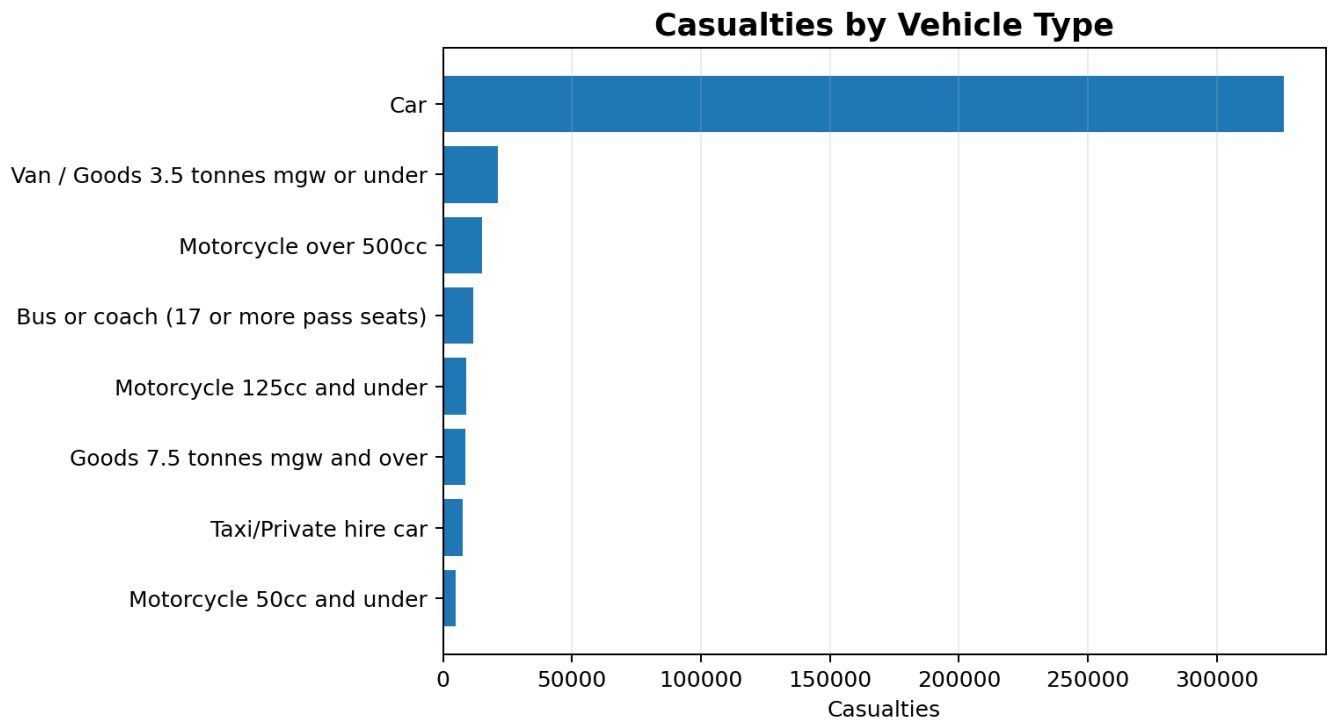
- Slight accidents account for the largest casualty volume: 351,436.
- Car is the largest vehicle category by casualties: 325,922.
- Single carriageway has the highest casualty count among road types: 309,698.
- Dry road conditions account for the highest casualty count: 279,445.
- Urban areas have more casualties than Rural areas in this dataset: 255,864.
- Daylight conditions have the highest casualty volume: 304,963.

2. Accident Severity Analysis



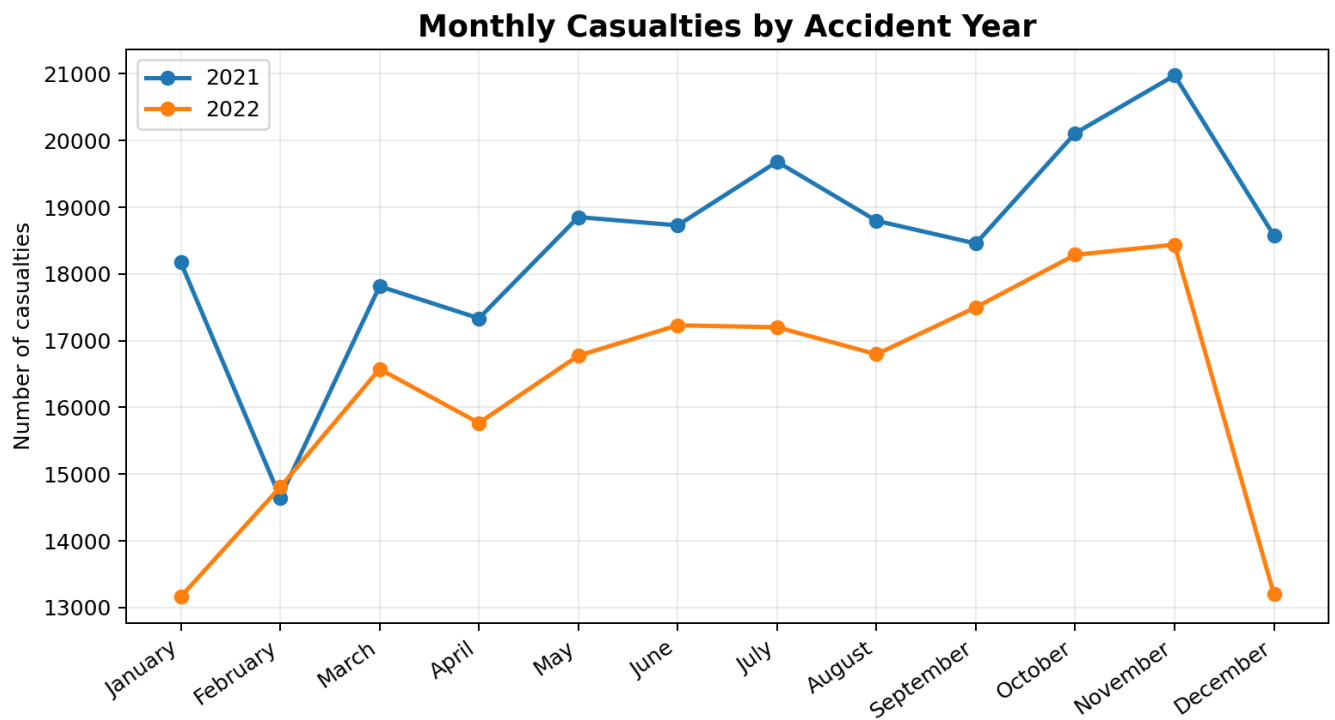
Severity is a major indicator of accident impact. The chart shows the distribution of casualties across fatal, serious and slight severity categories.

3. Vehicle Type Analysis



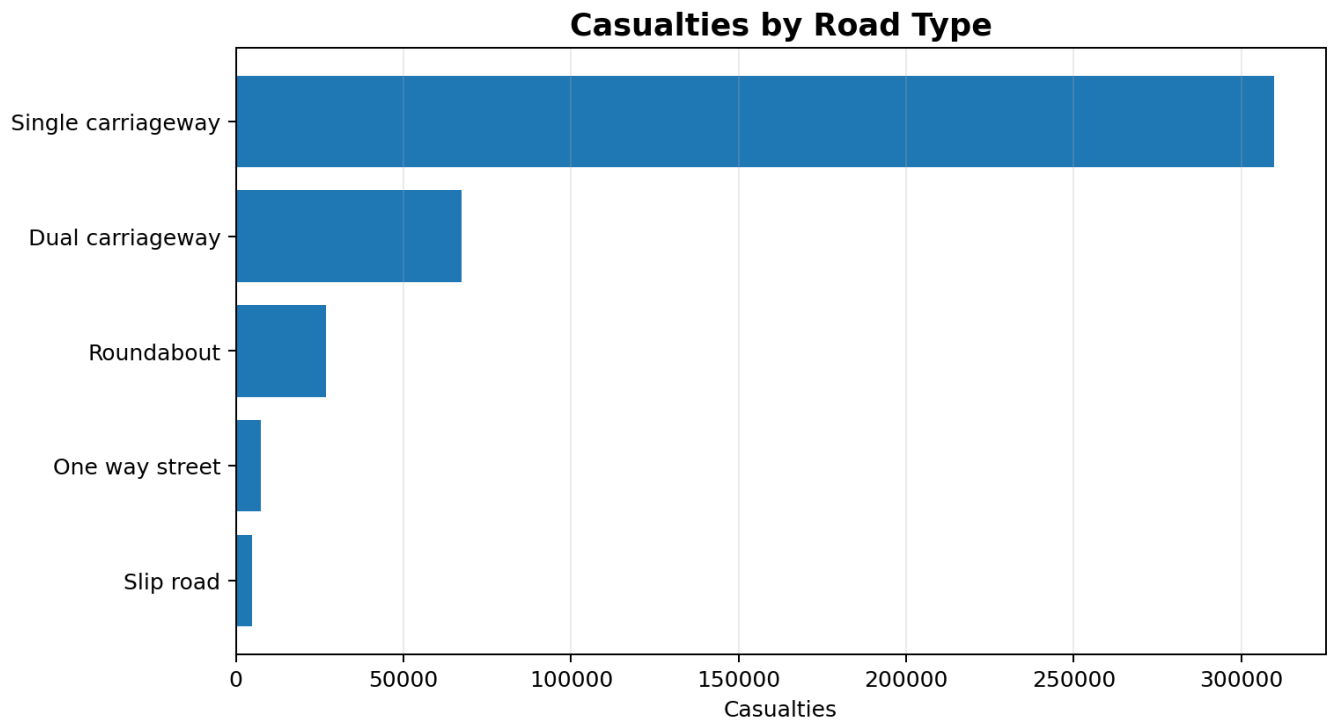
Cars contribute the largest share of casualties among the vehicle categories shown in the workbook.

4. Monthly Trend Analysis



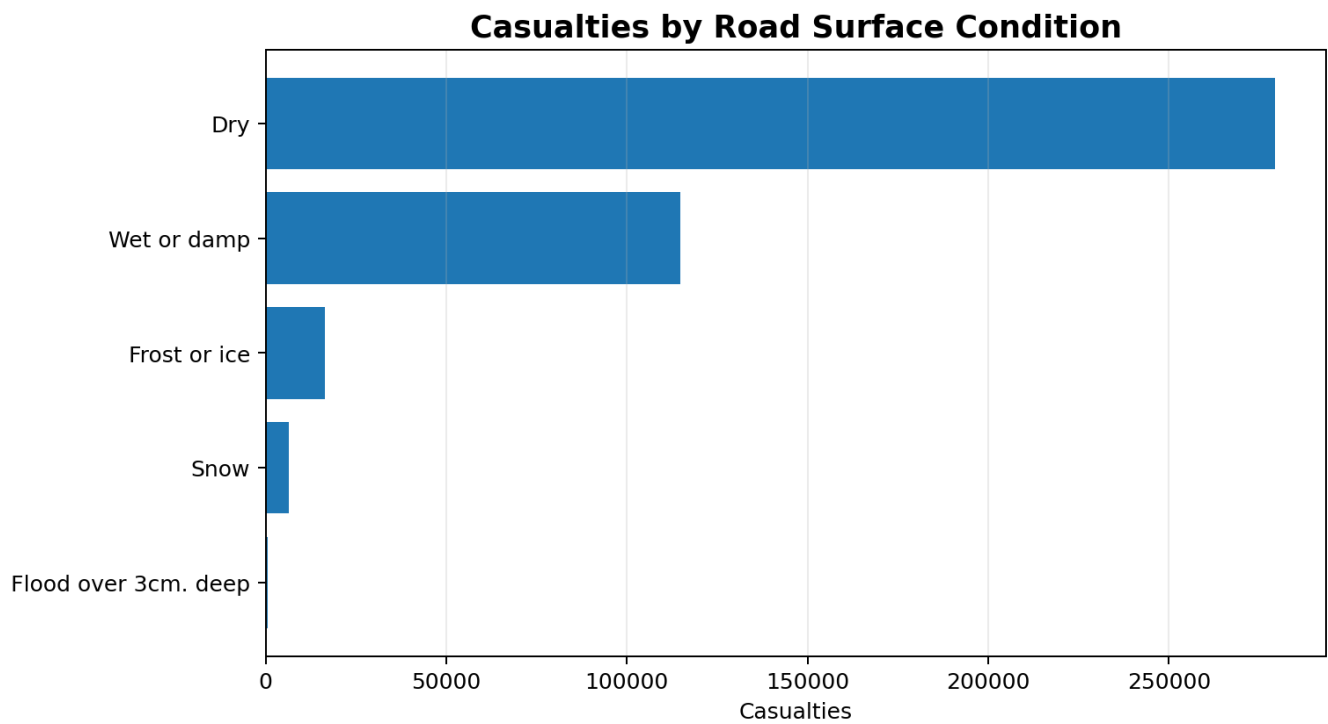
The monthly chart compares casualty volumes across the available accident years and helps identify seasonal or monthly peaks.

5. Road Type Analysis



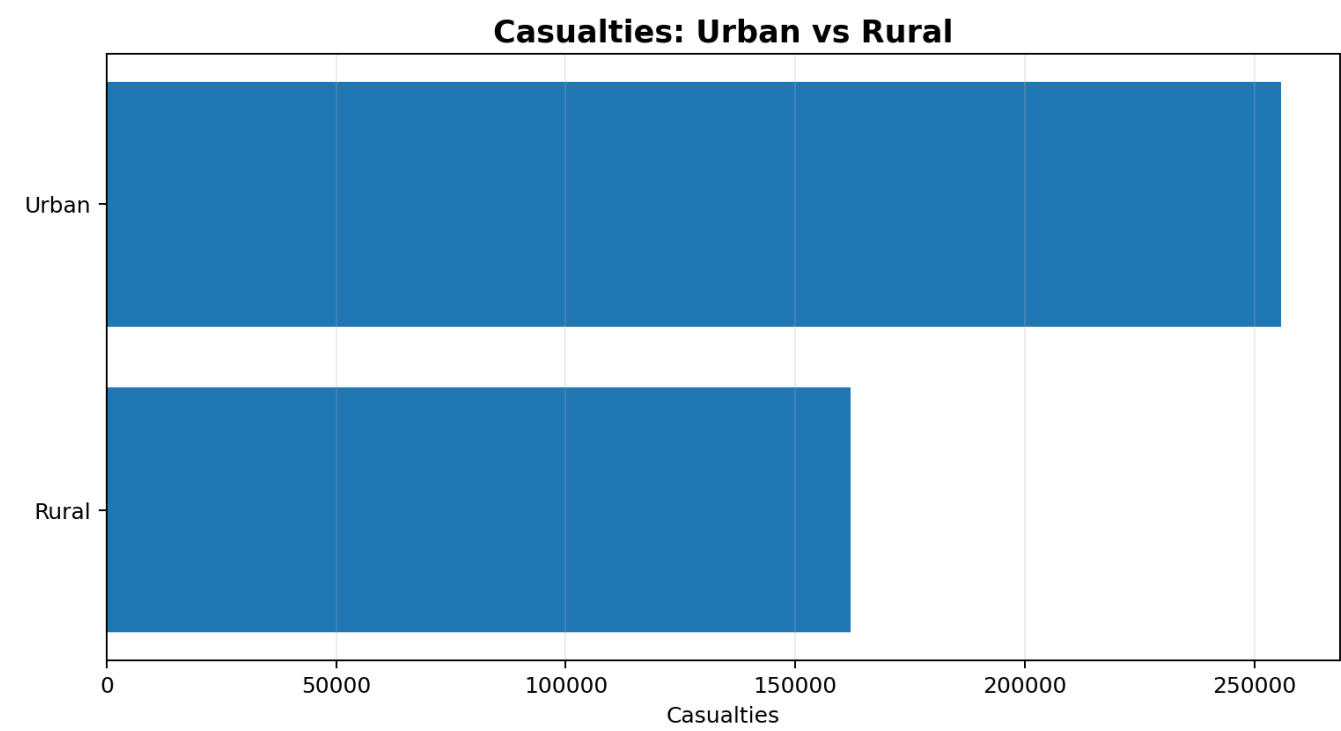
Single carriageways account for the largest casualty volume among the listed road types.

6. Road Surface Analysis



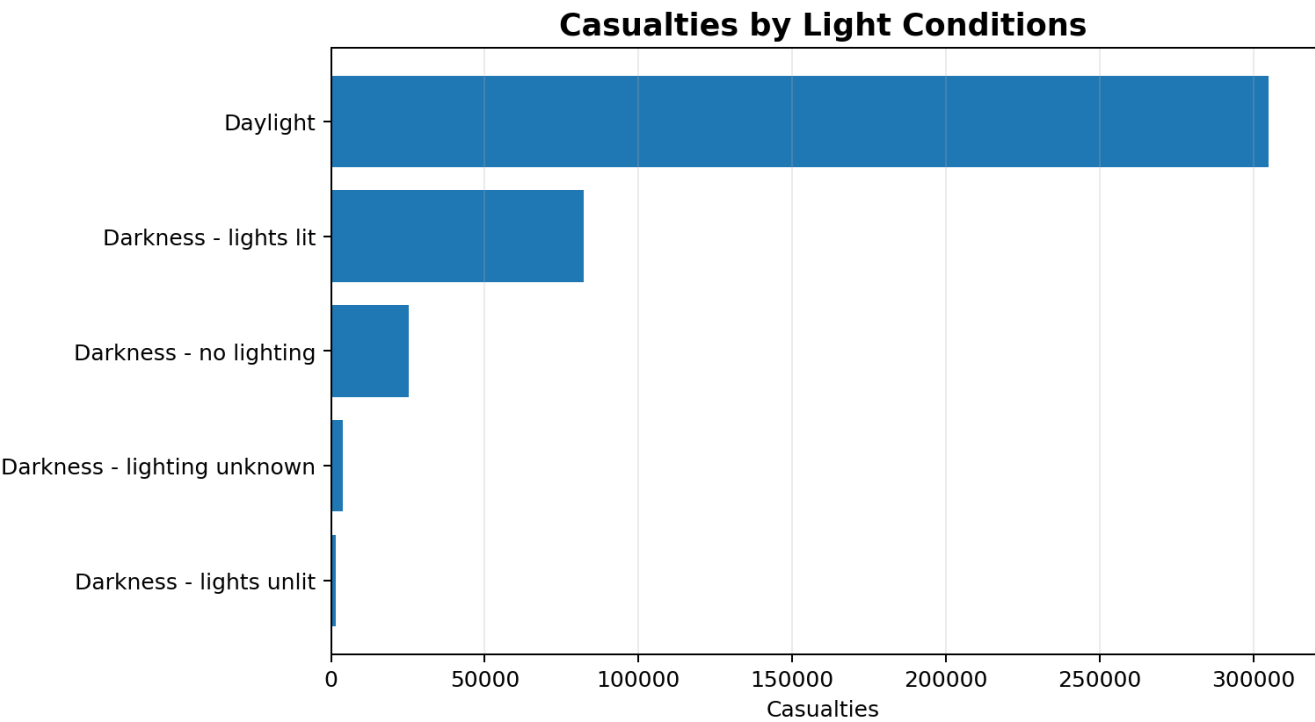
Dry road conditions have the highest casualty count in the dataset, followed by wet or damp conditions.

7. Urban vs Rural Analysis



The chart compares casualty concentration between urban and rural areas.

8. Light Conditions Analysis



Daylight conditions account for the largest casualty volume, while dark conditions represent a substantial secondary category.

9. Excel Analysis Workflow

Step	Technique	Purpose
1	Power Query	Clean, transform and prepare the raw accident data
2	Excel formulas	Support calculations and data checks
3	Pivot Tables	Summarize casualties across key dimensions
4	Pivot Charts	Visualize trends and category comparisons
5	Dashboard	Present KPIs, charts and interactive analysis

10. Conclusion

This project demonstrates a complete Excel analytics workflow: raw data preparation with Power Query, structured analysis with Pivot Tables, visualization through Pivot Charts, and dashboard-style reporting. The analysis can help users quickly identify high-volume casualty categories and focus further investigation on road type, vehicle type, surface condition, lighting and geographic/urban-rural patterns.

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